A Distributional Comparison Principle for Stochastic MPC Dynamic Regret

Corrected Partial Progress on the Inexact-Distribution Question

June 24, 2026

Abstract

Shin, Na, and Anitescu prove that, for finite-support linear-quadratic stochastic model predictive control (SMPC), the exact-distribution dynamic regret decays exponentially in the prediction horizon. They also identify dynamic regret under inexact distributional knowledge as an open direction. This note gives a corrected and explicit partial result. The main contribution is a comparison principle: if the exact-law regret theorem is available uniformly on a neighborhood of probability laws and the pathwise costs of the relevant policies are stable under distributional perturbations, then nominal SMPC designed for a law Q and evaluated under a true law P satisfies

$$J_P(\pi_Q^W) - J_P^\star \le B_T\rho^W + 2\omega(P,Q).$$

The note then verifies the stability term in two useful regimes. First, on a common finite scenario tree, the theorem of Shin–Na–Anitescu plus a positive lower bound on node probabilities gives a fully explicit total-variation bound

$$J_P(\pi_Q^W) - J_P^\star \le B_T\rho^W + 4M_T\,\mathrm{TV}(P,Q).$$

Second, a Wasserstein version follows for policy classes with uniformly Lipschitz closed-loop path costs; the state-recursion proof includes the missing initial-control term. The robust-ambiguity statement is deliberately separated as a conditional comparison theorem because proving the required robust exact-window estimate is essentially part of the remaining open problem.

This is therefore *partial progress*, not a complete solution of the distributionally robust dynamic-regret problem.

1 Status relative to the open problem

The exact-law result of Shin, Na, and Anitescu [1] assumes that the distribution of the uncertainty process is known when the conditional truncated SMPC problems are formed. Their Remark 5 and conclusion explicitly point out that dynamic regret under inexact distributional knowledge remains open. The purpose of this note is not to claim a complete solution. Instead, it records the part of the problem that follows from exact-law regret plus a distributional stability estimate, and it gives two rigorous verifications of that stability estimate.

The limitations are important:

- The common finite-tree theorem below does not handle support mismatch or histories that have zero nominal probability but positive true probability.
- The Wasserstein theorem requires the relevant SMPC optimizer maps to be uniformly bounded and Lipschitz functions of the disturbance path. That property is not proved here from the stabilizability/detectability assumptions of [1].

- The robust ambiguity-set theorem is conditional on a robust exact-window estimate. Such an estimate is not a formal consequence of the exact-law quadratic-program theorem, because worst-case ambiguity objectives need not retain the same smooth quadratic scenario-tree structure.

2 A path-space formulation

Fix a finite control horizon T. Let Ω be the set of disturbance paths $\xi = (\xi_0, \dots, \xi_T)$, equipped with a sigma-field. A nonanticipative policy π assigns controls $u_t^\pi(\xi_{0:t})$. For a fixed initial pair $w_{-1} = (x_{-1}, u_{-1})$, the state-control path is generated by

$$x_t = A(\xi_t)x_{t-1} + B(\xi_t)u_{t-1} + d(\xi_t), \qquad u_t = u_t^\pi(\xi_{0:t}). \tag{1}$$

The pathwise cost is denoted by $C^\pi(\xi)$. In the linear-quadratic setting,

$$C^\pi(\xi) = \sum_{t=0}^{T} \left\{ \frac{1}{2} \begin{bmatrix} x_t \\ u_t \end{bmatrix}^\top \begin{bmatrix} Q(\xi_t) & 0 \\ 0 & R(\xi_t) \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix} - \begin{bmatrix} q(\xi_t) \\ r(\xi_t) \end{bmatrix}^\top \begin{bmatrix} x_t \\ u_t \end{bmatrix} \right\}. \tag{2}$$

For a probability law P on Ω, define

$$J_P(\pi) := \mathbb{E}_{\xi \sim P}[C^\pi(\xi)], \qquad J_P^\star := \inf_\pi J_P(\pi).$$

When the W-step SMPC controller is computed using law Q, write π_Q^W for the resulting receding-horizon policy. In the exact-law case $Q = P$, this is the usual SMPC policy analyzed in [1]. In the misspecified case $Q \neq P$, $J_P(\pi_Q^W)$ is the true cost of the nominally designed controller.

3 The comparison principle

The first result is deliberately abstract. It isolates the only algebraic step that is needed to pass from exact-law regret to misspecified-law regret.

Assumption 3.1 (Uniform exact-law regret)**.** *There are constants $B_T < \infty$, $\rho \in (0, 1)$, and a class of probability laws $\mathcal{N}$ such that, for every $Q \in \mathcal{N}$ and every admissible prediction horizon W,*

$$0 \le J_Q(\pi_Q^W) - J_Q^\star \le B_T \rho^W. \tag{3}$$

Assumption 3.2 (Distributional stability for the relevant policies)**.** *There is a symmetric function $\omega : \mathcal{N} \times \mathcal{N} \to [0, \infty)$ such that, for every $P, Q \in \mathcal{N}$,*

$$\left| J_P(\pi_Q^W) - J_Q(\pi_Q^W) \right| \le \omega(P, Q) \tag{4}$$

for every relevant W, and for every $\varepsilon > 0$ there is an ε-optimal policy $\pi_{P,\varepsilon}^\star$ for $J_P^\star$ satisfying

$$\left| J_P(\pi_{P,\varepsilon}^\star) - J_Q(\pi_{P,\varepsilon}^\star) \right| \le \omega(P, Q). \tag{5}$$

Theorem 3.3 (Nominal comparison bound)**.** *Under Assumptions 3.1 and 3.2, for all $P, Q \in \mathcal{N}$,*

$$J_P(\pi_Q^W) - J_P^\star \le B_T \rho^W + 2\omega(P, Q). \tag{6}$$

Proof. Add and subtract the nominal-law cost and the nominal-law value:

$$J_P(\pi_Q^W) - J_P^\star = \big[J_P(\pi_Q^W) - J_Q(\pi_Q^W)\big] + \big[J_Q(\pi_Q^W) - J_Q^\star\big] + \big[J_Q^\star - J_P^\star\big].$$

The first bracket is at most $\omega(P,Q)$ by (4). The second bracket is at most $B_T\rho^W$ by (3). For the third bracket, let $\pi_{P,\varepsilon}^\star$ be as in Assumption 3.2. Then

$$J_Q^\star - J_P^\star \le J_Q(\pi_{P,\varepsilon}^\star) - J_P^\star \le J_Q(\pi_{P,\varepsilon}^\star) - J_P(\pi_{P,\varepsilon}^\star) + \varepsilon \le \omega(P,Q) + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the claim. □

Remark 3.4 (What is and is not proved)**.** *Theorem 3.3 is a theorem, but it is not by itself a solution of the open problem. The hard SMPC-specific work is hidden in two places: proving a uniform exact-law estimate on a distributional neighborhood, and proving a stability modulus for the closed-loop policies that are actually produced by the conditional receding-horizon problems.*

4 A finite common-support theorem

The next result is a concrete partial answer in the finite-support regime of [1]. It assumes a common scenario tree and excludes small nominal probabilities. This is a real restriction, but it removes the support/conditioning pathology and makes the comparison estimate rigorous.

Fix an initial realization $\xi_0 = \bar{\xi}_0$. Let $G = (V, E)$ be a finite scenario tree of depth T, and let V_t denote its nodes at stage t. Let Ω be the finite set of root-to-leaf paths. Every law below is understood as a law on this fixed tree, conditional on $\xi_0 = \bar{\xi}_0$. For a node $i \in V_t$, write $P(i)$ for the probability of reaching node i. Define

$$\mathrm{TV}(P,Q) := \sup_{A\subseteq\Omega} |P(A) - Q(A)| = \frac{1}{2}\sum_{\omega\in\Omega} |P(\omega) - Q(\omega)|\,.$$

Assumption 4.1 (Common finite tree with positive node masses)**.** *There is a number $\underline{\pi} > 0$ and a family $\mathcal{N}$ of laws on Ω such that every $P \in \mathcal{N}$ has the same tree support G and*

$$P(i) \ge \underline{\pi}, \qquad i \in V, \tag{7}$$

where probabilities are conditional on the fixed root $\xi_0 = \bar{\xi}_0$. The coefficient data on this tree satisfy Assumptions 1 and 2 of [1] with common constants $L \ge 1$, $\alpha \in (0,1)$, and $\gamma \in (0,1]$. Finally, with $p(\xi_t) := (q(\xi_t), r(\xi_t), d(\xi_t))$, let

$$\bar{p} := \max_{\omega\in\Omega}\max_{0\le t\le T} \|p(\omega_t)\| < \infty, \qquad \bar{g} := \max_{\omega\in\Omega}\max_{0\le t\le T} \|(q(\omega_t), r(\omega_t))\| < \infty. \tag{8}$$

Let $c_1, c_2, c_5, c_6, c_7, \rho$, and the minimum prediction length $\bar{W}$, be the constants appearing in Corollary 1 and Theorems 2–3 of [1], evaluated with the common constants in Assumption 4.1. Set

$$B_T := c_5\bar{p}^2 T + c_6\bar{p}\,\|w_{-1}\| + c_7\,\|w_{-1}\|^2\,. \tag{9}$$

For $r_\rho := \rho^{1/2}$, define

$$\begin{aligned}
Z_\star &:= \underline{\pi}^{-1/2} c_1 \left(2L\,\|w_{-1}\| + \frac{2\bar{p}}{1-\rho}\right),\\
Z_{\mathrm{mpc}} &:= \underline{\pi}^{-1/2} c_2 \left(2L\,\|w_{-1}\| + \frac{1+r_\rho}{1-r_\rho}\,\bar{p}\right),\\
Z &:= \max\{Z_\star, Z_{\mathrm{mpc}}\},\\
M_T &:= (T+1)\left(\frac{1}{2}LZ^2 + \bar{g}Z\right).
\end{aligned} \tag{10}$$

Lemma 4.2 (Uniform pathwise cost bound on the common tree)**.** *Under Assumption 4.1, for every $P \in \mathcal{N}$, every exact optimal policy $\pi_P^\star$, every $Q \in \mathcal{N}$, and every $W \geq \bar{W}$,*

$$\left| C^{\pi_P^\star}(\omega) \right| \leq M_T, \qquad \left| C^{\pi_Q^W}(\omega) \right| \leq M_T, \qquad \omega \in \Omega. \tag{11}$$

Proof. The EISSE bound for the exact optimal policy in Corollary 1 of [1] gives, for every $P \in \mathcal{N}$,

$$\left(\mathbb{E}_P \left[\|w_t^\star\|^2 \mid \xi_0 = \bar{\xi}_0 \right] \right)^{1/2} \leq c_1 \left(2L\rho^t \|w_{-1}\| + \frac{2}{1-\rho} D_P \right),$$

where $D_P := \max_t (\mathbb{E}_P[\|p(\xi_t)\|^2 \mid \xi_0 = \bar{\xi}_0])^{1/2} \leq \bar{p}$. Since each stage-$t$ node has probability at least $\underline{\pi}$, this expectation bound implies the pointwise node bound $\|w_t^\star(i)\| \leq Z_\star$ for all nodes $i \in V_t$.

The closed-loop SMPC perturbation bound, Theorem 2 of [1], similarly gives

$$\left(\mathbb{E}_Q \left[\left\| w_t^{\mathrm{cl,W}} \right\|^2 \mid \xi_0 = \bar{\xi}_0 \right] \right)^{1/2} \leq c_2 \left(2L\rho^{t/2} \|w_{-1}\| + \sum_{s=0}^{T} \rho^{|t-s|/2} D_Q \right).$$

Here $D_Q \leq \bar{p}$, and

$$\sum_{s=0}^{T} \rho^{|t-s|/2} \leq 1 + 2\sum_{k=1}^{\infty} r_\rho^k = \frac{1+r_\rho}{1-r_\rho}.$$

The lower node-mass bound therefore implies $\left\| w_t^{\mathrm{cl,W}}(i) \right\| \leq Z_{\mathrm{mpc}}$ on every node.

For either policy, $\|(x_t, u_t)\| \leq Z$. Because $Q(\xi_t)$ and $R(\xi_t)$ are L-bounded and $\|(q(\xi_t), r(\xi_t))\| \leq \bar{g}$, each stage cost has magnitude at most

$$\frac{1}{2} L Z^2 + \bar{g} Z.$$

Summing over $t = 0, \ldots, T$ proves (11). □

Theorem 4.3 (Nominal SMPC under finite-tree misspecification)**.** *Under Assumption 4.1, for every $P, Q \in \mathcal{N}$ and every $W \geq \bar{W}$,*

$$J_P(\pi_Q^W) - J_P^\star \leq B_T \rho^W + 4M_T \,\mathrm{TV}(P, Q), \tag{12}$$

with B_T and M_T defined in (9)–(10).

Proof. The exact-law theorem of [1], applied under law Q, gives

$$J_Q(\pi_Q^W) - J_Q^\star \leq B_T \rho^W,$$

because $D_Q \leq \bar{p}$. By Lemma 4.2, both $C^{\pi_Q^W}$ and $C^{\pi_P^\star}$ are bounded in absolute value by M_T. Therefore, for either of these policies π,

$$|J_P(\pi) - J_Q(\pi)| = \left| \sum_{\omega \in \Omega} (P(\omega) - Q(\omega)) C^\pi(\omega) \right| \leq M_T \sum_{\omega \in \Omega} |P(\omega) - Q(\omega)| = 2M_T \,\mathrm{TV}(P, Q).$$

The comparison proof of Theorem 3.3 with $\omega(P, Q) = 2M_T \,\mathrm{TV}(P, Q)$ yields (12). □

Remark 4.4 (Interpretation)**.** *Theorem 4.3 is a genuine finite-tree misspecification guarantee: the SMPC controller is computed with conditional probabilities from Q, but its cost is evaluated under P. The price of distributional misspecification is explicit. The theorem remains partial because the lower bound $\underline{\pi}$ rules out vanishing-probability histories, and the common-support assumption rules out nominal laws that fail to define conditional SMPC subproblems on histories that can occur under the true law.*

5 A Wasserstein version under pathwise Lipschitz policies

The previous theorem uses total variation on a finite tree. A Wasserstein statement is also immediate once pathwise costs are uniformly Lipschitz. This section corrects the common mistake of ignoring the fixed initial control u_{-1} in the state recursion.

Let (Ξ, d_Ξ) be a compact metric space and put

$$D_T(\xi,\zeta) := \sum_{s=0}^{T} d_\Xi(\xi_s,\zeta_s). \tag{13}$$

Assume the coefficient maps are bounded by M and Lipschitz:

$$\begin{gathered}\|A(z)\|, \|B(z)\|, \|d(z)\|, \|Q(z)\|, \|R(z)\|, \|q(z)\|, \|r(z)\| \le M,\\ \left\|A(z)-A(z')\right\| \le L_A d_\Xi(z,z'), \quad \left\|B(z)-B(z')\right\| \le L_B d_\Xi(z,z'), \quad \left\|d(z)-d(z')\right\| \le L_d d_\Xi(z,z'),\end{gathered} \tag{14}$$

and similarly with constants L_Q, L_R, L_q, L_r for Q, R, q, r. Let $\mathcal{C}$ be a class of nonanticipative policies such that, for every $\pi \in \mathcal{C}$,

$$\|u_t^\pi(\xi_{0:t})\| \le U, \qquad \|u_t^\pi(\xi_{0:t}) - u_t^\pi(\zeta_{0:t})\| \le K \sum_{s=0}^{t} d_\Xi(\xi_s,\zeta_s). \tag{15}$$

Define

$$U_{-1} := \|u_{-1}\|, \qquad U_s := U \ (s\ge 0), \qquad K_{-1} := 0, \qquad K_s := K \ (s \ge 0).$$

Set

$$S_{-1} := \|x_{-1}\|, \qquad S_t := MS_{t-1} + MU_{t-1} + M, \tag{16}$$

$$H_{-1} := 0, \qquad H_t := MH_{t-1} + MK_{t-1} + L_A S_{t-1} + L_B U_{t-1} + L_d. \tag{17}$$

For $t = 0, \dots, T$, define

$$\begin{aligned}\Gamma_t := {} & \frac{1}{2} L_Q S_t^2 + M S_t H_t + \frac{1}{2} L_R U^2 + MUK \\ & + L_q S_t + M H_t + L_r U + MK, \qquad \kappa_T := \sum_{t=0}^{T} \Gamma_t.\end{aligned} \tag{18}$$

Proposition 5.1 (Pathwise Lipschitz cost bound)**.** *Under (14)–(15), every $\pi \in \mathcal{C}$ satisfies*

$$\|x_t^\pi(\xi)\| \le S_t, \qquad \|x_t^\pi(\xi) - x_t^\pi(\zeta)\| \le H_t D_T(\xi,\zeta), \tag{19}$$

and its pathwise cost is κ_T-Lipschitz:

$$|C^\pi(\xi) - C^\pi(\zeta)| \le \kappa_T D_T(\xi,\zeta). \tag{20}$$

Consequently, for any laws P, Q on Ξ^{T+1},

$$|J_P(\pi) - J_Q(\pi)| \le \kappa_T W_{1,D_T}(P,Q), \tag{21}$$

where the Wasserstein distance is computed with ground metric D_T.

Proof. The state-size bound follows from (1):

$$\|x_t(\xi)\| \le M\,\|x_{t-1}(\xi)\| + M\,\|u_{t-1}(\xi)\| + M \le MS_{t-1} + MU_{t-1} + M = S_t.$$

For the Lipschitz bound, subtract the two state recursions. Since u_{-1} is fixed, its Lipschitz constant is $K_{-1} = 0$. For $t \ge 0$,

$$\begin{aligned}\|x_t(\xi) - x_t(\zeta)\| &\le M\,\|x_{t-1}(\xi) - x_{t-1}(\zeta)\| + M\,\|u_{t-1}(\xi) - u_{t-1}(\zeta)\| \\ &\quad + L_A S_{t-1} d_\Xi(\xi_t,\zeta_t) + L_B U_{t-1} d_\Xi(\xi_t,\zeta_t) + L_d d_\Xi(\xi_t,\zeta_t) \\ &\le \big(MH_{t-1} + MK_{t-1} + L_A S_{t-1} + L_B U_{t-1} + L_d\big) D_T(\xi,\zeta),\end{aligned}$$

which is (17).

It remains to bound the stage cost. For the state quadratic term,

$$\begin{aligned}&\Big| x_t(\xi)^\top Q(\xi_t) x_t(\xi) - x_t(\zeta)^\top Q(\zeta_t) x_t(\zeta) \Big| \\ &\qquad \le L_Q S_t^2 D_T(\xi,\zeta) + 2MS_tH_tD_T(\xi,\zeta).\end{aligned}$$

After multiplying by $1/2$, this gives the first two terms of Γ_t. The control quadratic term is bounded in the same way by

$$\left(\frac{1}{2}L_R U^2 + MUK\right) D_T(\xi,\zeta),$$

and the linear state and control terms are bounded by

$$(L_q S_t + MH_t) D_T(\xi,\zeta), \qquad (L_r U + MK) D_T(\xi,\zeta).$$

Summing over t proves (20). The Wasserstein estimate (21) follows from the Kantorovich–Rubinstein duality. □

Corollary 5.2 (Wasserstein comparison, conditional on Lipschitz SMPC maps)**.** *Suppose Assumption 3.1 holds on a class $\mathcal{N}$ of laws on Ξ^{T+1}, and suppose that every policy needed in Assumption 3.2 belongs to a class $\mathcal{C}$ satisfying (15). Then*

$$J_P(\pi_Q^W) - J_P^\star \le B_T \rho^W + 2\kappa_T W_{1,D_T}(P,Q). \tag{22}$$

Proof. Proposition 5.1 gives Assumption 3.2 with $\omega(P,Q) = \kappa_T W_{1,D_T}(P,Q)$. Theorem 3.3 gives the claim. □

Remark 5.3 (The remaining optimizer-regularity gap)**.** *Corollary 5.2 should not be read as proving that the SMPC optimizer maps of [1] are uniformly Lipschitz in the disturbance path. It proves the Wasserstein regret bound once that closed-loop Lipschitz property is available. Establishing such a property from stabilizability, detectability, and the scenario-tree KKT perturbation theory is a separate and nontrivial task.*

6 Robust ambiguity sets: a conditional comparison only

For an ambiguity set $\mathcal{P} \subseteq \mathcal{N}$, define

$$\overline{J}_{\mathcal{P}}(\pi) := \sup_{Q\in\mathcal{P}} J_Q(\pi), \qquad \overline{J}^\star_{\mathcal{P}} := \inf_{\pi} \overline{J}_{\mathcal{P}}(\pi).$$

Let $\pi_{\mathcal{P}}^W$ be a candidate robust receding-horizon policy. The next theorem is included to clarify what can be proved by comparison alone.

Assumption 6.1 (Robust exact-window estimate)**.** *For constants $B_T < \infty$ and $\rho \in (0,1)$,*

$$\overline{J}_{\mathcal{P}}(\pi^W_{\mathcal{P}}) - \overline{J}^\star_{\mathcal{P}} \le B_T \rho^W. \tag{23}$$

Assumption 6.2 (Stability on the ambiguity set)**.** *There is $\kappa < \infty$ and a metric d such that, for every $P, Q \in \mathcal{P}$, for $\pi = \pi^W_{\mathcal{P}}$, and for ε-optimal true-law policies,*

$$|J_P(\pi) - J_Q(\pi)| \le \kappa \mathsf{d}(P, Q).$$

Theorem 6.3 (Conditional robust comparison)**.** *Under Assumptions 6.1 and 6.2, for every true law $P \in \mathcal{P}$,*

$$J_P(\pi^W_{\mathcal{P}}) - J^\star_P \le B_T \rho^W + \kappa \operatorname{diam}_{\mathsf{d}}(\mathcal{P}). \tag{24}$$

In particular, if $\mathcal{P} = \{Q : \mathsf{d}(Q, \widehat{P}) \le \varepsilon\}$, then the second term is at most $2\kappa\varepsilon$.

Proof. Since $P \in \mathcal{P}$,

$$J_P(\pi^W_{\mathcal{P}}) \le \overline{J}_{\mathcal{P}}(\pi^W_{\mathcal{P}}) \le \overline{J}^\star_{\mathcal{P}} + B_T \rho^W.$$

Let $\pi^\star_{P,\varepsilon}$ be ε-optimal for $J^\star_P$. Then

$$\begin{aligned} \overline{J}^\star_{\mathcal{P}} \le \overline{J}_{\mathcal{P}}(\pi^\star_{P,\varepsilon}) &= \sup_{Q \in \mathcal{P}} J_Q(\pi^\star_{P,\varepsilon}) \\ &\le J_P(\pi^\star_{P,\varepsilon}) + \kappa \operatorname{diam}_{\mathsf{d}}(\mathcal{P}) \le J^\star_P + \varepsilon + \kappa \operatorname{diam}_{\mathsf{d}}(\mathcal{P}). \end{aligned}$$

Let $\varepsilon \downarrow 0$. The ball statement follows from $\operatorname{diam}(\mathcal{P}) \le 2\varepsilon$. $\square$

Remark 6.4 (Why this is not the robust solution)**.** *Assumption 6.1 is the missing robust dynamic-regret theorem. It cannot simply be imported from the exact-law quadratic theorem: a worst-case ambiguity objective may be nonsmooth, may require rectangularity or a nested distance to behave dynamically, and may not have the same scenario-tree KKT matrix structure. Theorem 6.3 is therefore a useful reduction, not a solution.*

7 A one-stage lower bound

The distributional-error term cannot be removed in general. Consider the one-stage scalar decision problem

$$\ell(u; \xi) = \frac{\eta}{2} u^2 - \xi u, \qquad \eta > 0, \qquad \xi \in \{-1, 1\}.$$

Let

$$P_\delta(\xi = 1) = \frac{1+\delta}{2}, \qquad Q_\delta(\xi = 1) = \frac{1-\delta}{2}, \qquad \delta \in (0,1).$$

With ground distance $d_\Xi(-1, 1) = 1$, $W_1(P_\delta, Q_\delta) = \delta$. The means are $\mathbb{E}_{P_\delta}\xi = \delta$ and $\mathbb{E}_{Q_\delta}\xi = -\delta$, so

$$u^\star_{P_\delta} = \frac{\delta}{\eta}, \qquad u^\star_{Q_\delta} = -\frac{\delta}{\eta}.$$

Under the true law P_δ, the regret of the nominally optimal action is

$$\begin{aligned} J_{P_\delta}(u^\star_{Q_\delta}) - J_{P_\delta}(u^\star_{P_\delta}) &= \left(\frac{\eta}{2}\frac{\delta^2}{\eta^2} + \frac{\delta^2}{\eta}\right) - \left(\frac{\eta}{2}\frac{\delta^2}{\eta^2} - \frac{\delta^2}{\eta}\right) \\ &= \frac{2\delta^2}{\eta}. \end{aligned}$$

For fixed conditioning η, this is a quadratic local perturbation. Uniformly over η, no such quadratic behavior is possible: choosing $\eta = \delta$ gives regret $2\delta = 2W_1(P_\delta, Q_\delta)$. Thus a regret bound depending only on the prediction window, and not on distributional error, is impossible even in a one-stage problem.

8 Conclusion

The corrected result is best described as follows.

> *Exact-law SMPC dynamic regret plus distributional stability implies misspecified-law dynamic regret. On a common finite scenario tree with probabilities bounded away from zero, this gives an explicit total-variation partial solution. Wasserstein bounds follow under an additional closed-loop pathwise Lipschitz assumption. The full distributionally robust dynamic-regret problem remains open because the robust exact-window estimate and support/conditioning issues are not settled here.*

References

[1] S. Shin, S. Na, and M. Anitescu. Near-optimal performance of stochastic model predictive control. *arXiv:2210.08599v3*, 2025. https://arxiv.org/abs/2210.08599.

[2] C. Villani. *Optimal Transport: Old and New*. Springer, 2009.

[3] G. Ch. Pflug and A. Pichler. *Multistage Stochastic Optimization*. Springer, 2014.